Facebook Social Network Analysis Report

Dataset source:
SNAP Facebook Dataset

File used for analysis:

facebook_combined.txt

Network Statistics

After importing the dataset into Gephi, the following statistics were observed:

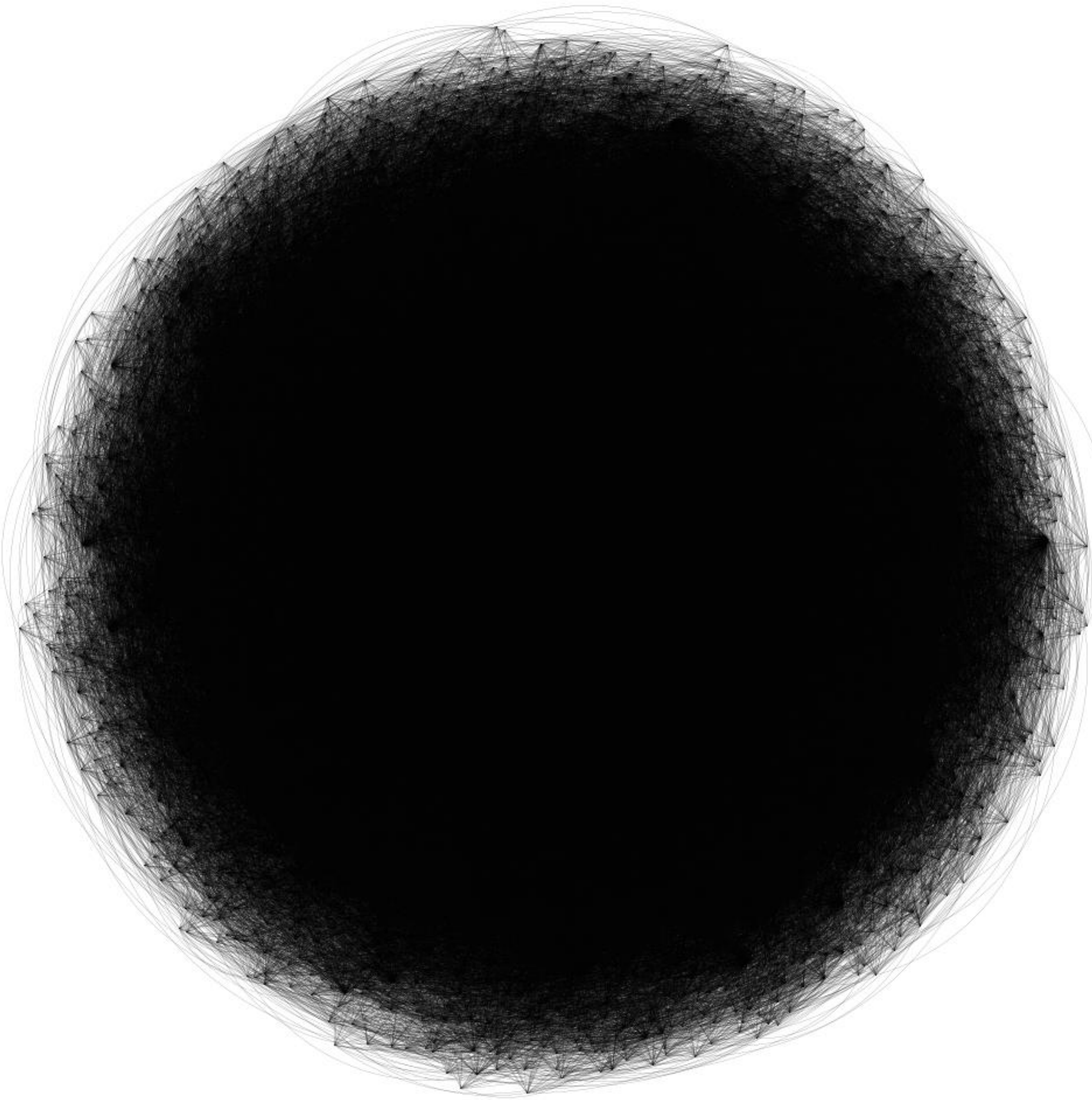
Metric	Value
Number of Nodes	4,039
Number of Edges	88,234
Graph Type	Undirected
Dataset Type	Social Network

Network Visualization

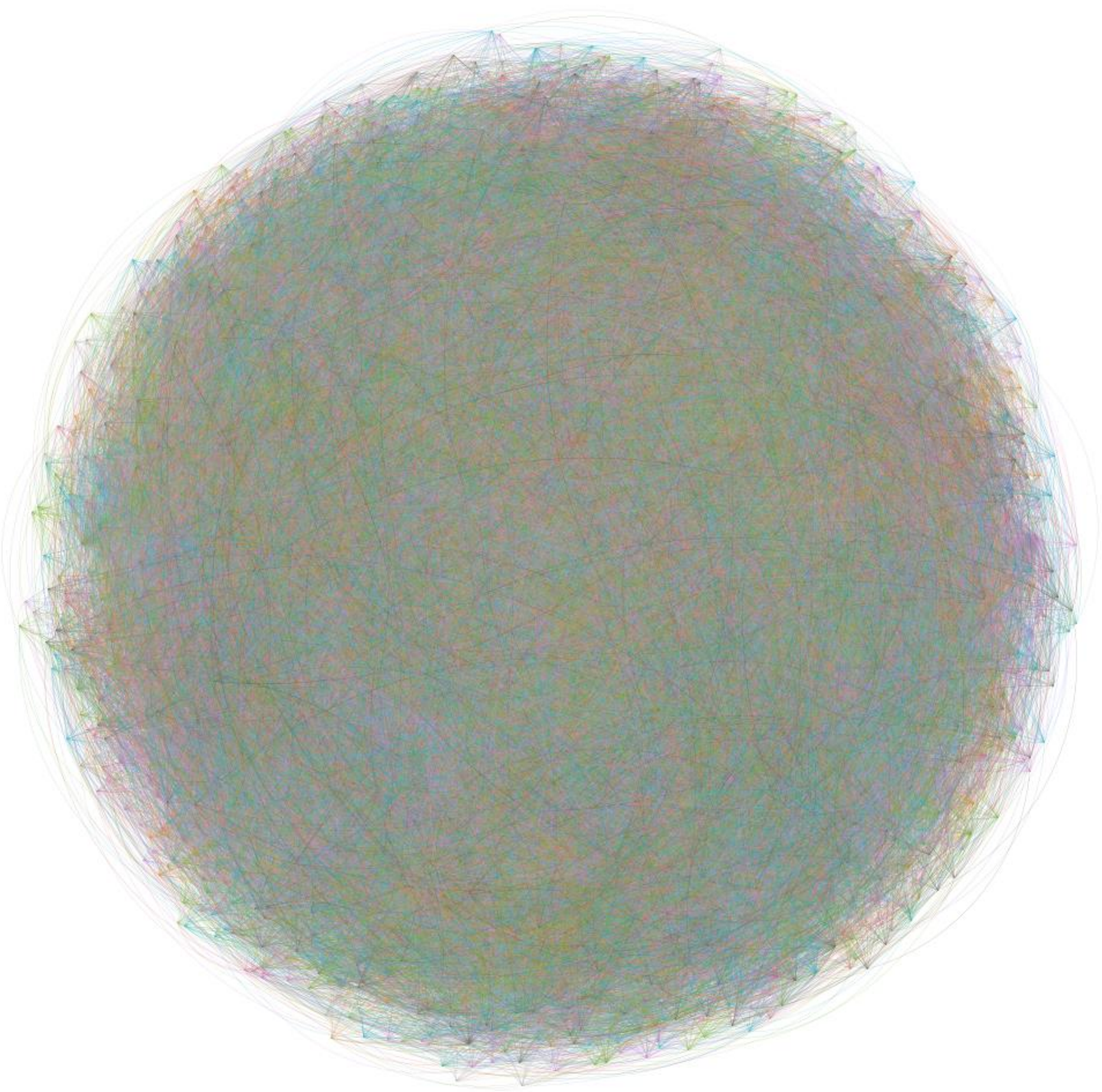
The network was visualized in Gephi using the ForceAtlas2 layout algorithm. This layout distributed nodes in a way that clearly revealed clusters and relationships among users.

Two visualizations were generated:

1. Full Network Visualization



2. Community (Modularity) Visualization



The modularity visualization highlighted different social communities within the Facebook network.

Degree Centrality Analysis

Degree centrality measures the number of direct connections a node has in the network. Nodes with high degree values are considered highly influential because they are connected to many other users.

Top 5 Nodes by Degree

Rank	Node ID	Degree
1	25	522
2	30	511
3	27	484

Rank	Node ID	Degree
4	28	459
5	0	451

These nodes act as major hubs in the network and maintain strong connectivity among users.

Betweenness Centrality Analysis

Betweenness centrality identifies nodes that act as bridges between different communities. Nodes with high betweenness centrality help connect separate clusters and support information flow across the network.

Top 5 Nodes by Betweenness Centrality

Rank	Node ID	Betweenness Centrality
1	25	0.025156730154955172
2	30	0.024167974403489865
3	27	0.021334883786245748
4	28	0.020174813254607016
5	0	0.019539495373458896

These nodes are important bridge nodes because they connect different parts of the social network. Removing such nodes could reduce communication efficiency between communities.

Degree Filter Analysis

A degree filter was applied using the following condition:

Degree >= 5

The filter removed weakly connected nodes and highlighted the core structure of the Facebook network.

Filter Results

Metric	Original Network	Filtered Network
Nodes	4,039	4,025
Edges	88,234	82,471
Visible Nodes	100%	99.65%
Visible Edges	100%	93.32%

The filtered network retained most of the important nodes and edges while removing less connected nodes. This made the network visualization cleaner and improved the visibility of major communities and influential users.

Comparison Between Original and Filtered Network

Original Network	**Filtered Network**
More cluttered visualization	Cleaner structure
Includes weakly connected nodes	Highlights influential nodes
Communities less visible	Communities easier to identify

The filtered network provided a better understanding of the central structure of the Facebook social graph and helped emphasize important users and communities.

Interpretation of the Network Structure

The Facebook network demonstrates characteristics commonly found in real-world social networks:

- A small number of users have very high connectivity.
- Highly connected hub nodes maintain the overall structure of the network.
- Communities naturally emerge among groups of related users.
- Bridge nodes help connect different communities and enable communication across the network.

The presence of hub nodes indicates a scale-free network structure, which is common in online social platforms.

Learning Outcomes

Through this analysis, the following concepts and techniques were learned using Gephi:

- Importing and visualizing social network datasets
- Applying graph layout algorithms such as ForceAtlas2
- Calculating degree centrality and betweenness centrality
- Identifying influential and bridge nodes
- Detecting communities using modularity
- Filtering networks based on node degree
- Understanding the structure and behavior of social networks

This project provided practical experience in network analysis, visualization, and interpretation using Gephi.